

Electronic design automation (EDA) plays the most important role in design and development of any product. Pravin Madhani, general manager, place and route division of Mentor Graphics, also the founder of Sierra Design Automation and Everest Design Automation, spoke to Shweta Dhadiwal of EFY Bureau on what leads to new tool development in the EDA industry. During the talk at EDA Tech Forum, he also expressed his views on Open standards and Open Source



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“For anything to have wide adoption, Open standards are must”

What are the key drivers for innovation in EDA tools?

The motivation behind any development in the EDA industry is automation. The objective is to make designers' life easier and simpler by providing tools that will automate the development process.

As chip designers innovate to go to lower process nodes like 22nm and 15nm, new challenges show up. Every time you go to a lower process node, you get a new process fix for a new design as you get new requirements related to power, real estate, etc. All these challenges need to be addressed simultaneously with a new set of tools.

What leads to the success of a new EDA tool?

Tools are developed for automating the development process. But time-to-market and verification are the major features that will make a tool successful. A tool can automate the work so that you don't have to do it manually, but if it takes a long time to process then what is its use? Even if the tool is automated, it has to be fast enough to speed up your development process.

Verification is another important characteristic. It helps to detect the error, capture the instance of its occurrence (to analyse and make sure that it doesn't reappear) and eliminate it before production.

The EDA industry works aggressively on innovative tool algorithms, tool intelligence, etc to stay a step ahead of the designers to make sure that their development is smooth, fast and flawless.

Which segment is the most difficult to handle?

There is no one particular segment. You need error-free development, but at the same time you need to complete your project in time. EDA companies have teams working in different domains like multi-threading and user interface, and developing expertise in that domain. Having a varied group of people in an EDA organisation helps to answer all the demands from designers.

What are the top five technologies that will support the EDA industry in a big way?